

A Hybrid Approach to Mental Health Classification: Integrating Topic Modeling and BERT for Sentiment Analysis

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Abstract

This study explores hybrid approaches to mental health text classification by integrating unsupervised topic modeling with supervised deep learning. We developed classifiers for seven mental health categories (e.g., Normal, Bipolar, Suicidal) using LDA, K-Means clustering, and BERT embeddings, as well as combined hybrid architectures. The results demonstrate that deep contextual embeddings are the primary driver of performance; the BERT baseline achieved 70% accuracy, vastly outperforming LDA (34%) and K-Means (36%). Furthermore, while the unsupervised methods suffered from class collapse on lower-frequency labels, the addition of BERT embeddings corrected these errors. However, the hybrid models failed to significantly outperform the BERT-only baseline, suggesting that traditional topic features provide minimal discriminative signal beyond what is already captured by transformer attention mechanisms in this domain. The study offers insights into balancing predictive performance and interpretability in mental-health NLP applications.

1 Introduction

1.1 Motivation

The growing mental health crisis among adolescents and young adults has intensified the need for reliable, scalable tools capable of identifying early

signs of distress. Recent national surveys highlight the scope of this issue: large numbers of 14 to 22-year-olds experiencing moderate to severe depressive symptoms are actively turning to the internet for mental health support, with 90% researching mental health issues online, 75% accessing others’ personal stories, 38% using well-being-related apps, and 32% connecting with providers through digital tools [2]. These findings, which were derived from a nationally representative sample of over 1,300 US teens and young adults using the validated PHQ-8 depression scale, underscore how deeply digital platforms have become embedded in young people’s mental health journeys. Additional research shows that online environments can help individuals overcome social interaction barriers, as social media does not require the immediate responses necessary in face-to-face communication [3]. This makes it a valuable resource for those whose symptoms hinder in-person interaction. Moreover, many teens report that online communities offer connection, support, and inspiration, particularly during periods of depression, stress, or anxiety [2].

Given this heavy reliance on digital spaces for emotional expression and support, there is a pressing need for computational methods capable of interpreting the nuanced, context-rich language people use when discussing mental health online. Traditional sentimental analysis tools often fail to capture the complexity of mental health discourse, while black-box deep learning models may lack the interpretability required in sensitive, high-stakes situations.

1.2 Problem Statement

The dataset used in this study contains user-generated mental-health-related texts labeled across seven categories: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, and Personality Disorder. Accurately distinguishing between these categories is challenging because the language associated with different mental states often overlaps, varies in intensity, and contains subtle emotional cues.

As such, the problem addressed in this research is the development of a system capable of reliably classifying text into these seven mental health categories to support early identification, monitoring, and mental health understanding. In other words, our goal is to develop a model that provides accurate classification of mental-health-related text while preserving interpretability for sensitive, clinically important applications.

1.3 Proposed Approach

To address the challenge of classifying mental health related text into seven psychological categories, this project explores two complementary hybrid modeling strategies. Both approaches combine unsupervised structure discovery with deep contextual representation learning, followed by a logistic regression classifier as a lightweight, interpretable prediction layer.

The first approach integrates Latent Dirichlet Allocation (LDA), an unsupervised topic-modeling technique, with BERT-based sentence embeddings. LDA identifies latent themes and emotional patterns present in the text, to provide an interpretable topic distribution for each document. These topic vectors are then concatenated with BERT embeddings, which capture rich semantic and contextual information. A logistic regression classifier is trained on the combined feature space to evaluate whether topic awareness can enhance predictive accuracy or interpretability.

The second approach replaces LDA with a clustering-based representation, capturing unsupervised structure by applying K-Means clustering to TF-IDF representations of the text. This second approach was motivated by the intuition that clustering lexical patterns may perform better on the many short, sparse posts in the dataset, where topic models like LDA often struggle. K-Means is fitted on TF-IDF vectors to identify groups of documents with similar lexical patterns. Each document’s assigned cluster label (or one-hot/encoded form) is then appended to its BERT embedding, forming another hybrid feature vector. Logistic regression is again used as the classifier, allowing us to assess whether clustering-based structure, learned from traditional vector space models, adds complementary information beyond BERT alone.

To rigorously evaluate the contribution of each component, we first establish independent baselines for the unsupervised methods (LDA and K-Means) and the supervised deep learning method (BERT). This benchmarking process allows us to quantify exactly how much predictive power is derived from semantic embeddings versus thematic or geometric structure.

Together, these two approaches provide a systematic comparison between topic-based structure (LDA) and cluster-based structure (K-Means) and how each interacts with deep contextual embeddings in the task of mental health text classification.

2 Methodology

2.1 Dataset

We utilized a dataset from Kaggle [1] that compiles various mental health statuses and statements from multiple Kaggle datasets, which in turn, collected the data from a variety of sources (like social media posts). Each statement within the dataset is tagged with one of seven mental health statuses: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, and Personality Disorder. The features of the dataset include a unique_id for each entry, the Statement reflecting an individual’s mental health status, and the corresponding Mental Health Status tag.

Prior to model training, we performed a series of pre-processing steps to enhance the dataset’s suitability for analysis. For text cleaning, we:

- Converted all text to lowercase to maintain uniformity;
- Removed irrelevant URLs, emojis, and punctuation to focus on substantive content;
- Expanded contractions (e.g., “don’t” becomes “do not”) to ensure clarity and improve the model’s understanding of the textual data;

Following this, we used WordNetLemmatizer from Python’s NLTK library to lemmatize the words in the dataset. This process involved converting words to their base or dictionary form (e.g., “running” to “run” and “better” to “good”), which reduces inflected forms and helps standardize the vocabulary used in the analysis. By lemmatizing the text, we aimed to improve the model’s accuracy in understanding the underlying meanings of words and their contextual use.

2.2 LDA for Topic Modeling

Latent Dirichlet Allocation (LDA) is a topic-modeling algorithm that discovers hidden themes (topics) in a large collection of text, which is done by looking at patterns in how words appear together. Before training an LDA model, the text is typically converted into a bag-of-words representation, where each document is expressed as a vector of word counts. This representation ignores grammar and word order but preserves word frequency

information, which is what LDA relies on to detect co-occurrence patterns. The general steps for how LDA works are as follows:

1. Initialize model parameters: Specify the number of topics, the number of iterations/passes, and the random seed (to ensure reproducibility).
2. Randomly assign each word in each document to a topic.
3. Examine two relationships:
 - a. Topic-Word: How often a word appears in a topic across the entire dataset;
 - b. Document-Topic: How often a topic appears in the document.
4. Using the counts of the two relationships that were examined in (2), update the topic assignment for each word.
5. Repeat steps 3 and 4 for the number of iterations or passes specified. The algorithm should continue iterating until the model stabilizes.

LDA produces two outputs:

1. *Set of topics*: Each topic is a list of high-probability words.
2. *Topic proportions for each document*: For each document, there is a vector that shows the proportion of each topic that appears in that document.

In our approach, we trained an LDA model using Gensim, specifying 7 topics to align with the seven mental-health categories in our dataset. However, the number of topics is not fixed and can be tuned depending on the size of the dataset and the level of granularity desired. For example, we could have experimented with 5–10 topics; fewer topics may produce broader, more general themes, while more topics may capture finer-grained distinctions between mental-health states. The model was trained for 10 passes over the corpus, a value that balances computational efficiency and topic stability. A fixed random state (42) was used to ensure reproducibility across runs.

2.3 Baseline Model 1: LDA Only

We developed a first baseline model that relies exclusively on probabilistic topic modeling features. By using LDA as a standalone classifier input, we aim to determine the extent to which pure thematic content, without the deep semantic context provided by transformers, can discriminate between mental health conditions.

2.3.1 Topic Feature Extraction

Feature extraction was performed by training an LDA model on the corpus to discover latent topic structures. For every statement in the dataset, the LDA model inferred a topic distribution vector, θ_d , where each element represents the probability of the document belonging to a specific topic. These probability vectors constitute the feature set X for the classifier. Unlike BERT embeddings, which capture semantic meaning and sentence structure, these features strictly represent the thematic composition of the text.

2.3.2 Training the Classifier

We trained a multiclass Logistic Regression classifier using the generated LDA topic distributions. The data was split into training and testing sets using an 80-20 ratio with stratified sampling to preserve the class distribution of the seven mental health categories.

Given the reliance on purely thematic features, we anticipated challenges in separating semantically complex categories (e.g., distinguishing *Depression* from *Suicidal* intent based solely on topic keywords). To address the inherent class imbalance observed in categories such as *Personality Disorder* and *Stress*, the classifier was configured with balanced class weights. This adjustment automatically scales the penalty for misclassifying distinct classes inversely proportional to their frequencies in the input data. The model was trained with a maximum of 5,000 iterations to ensure convergence on the lower-dimensional probabilistic feature space.

2.4 Baseline Model 2: BERT Only

For our baseline model, we developed a model that uses BERT embeddings as the sole feature representation for classifying mental health statements. BERT (Bidirectional Encoder Representations from Transformers) is an NLP

model developed by Google, which learns deep bidirectional representations of text [4]. Unlike sequential models such as RNNs, BERT leverages a transformer architecture that processes words in relation to all other words in a sentence using attention mechanisms, allowing it to capture richer contextual information. This enables more accurate representations of sentences, which can then be applied to a wide range of tasks, including sentiment and mental health classification [5].

2.4.1 Embedding Generation

We first generated embeddings for each statement in our dataset using the all-MiniLM-L6-v2 model from the SentenceTransformers library. Each statement was converted into a 384-dimensional vector representing its semantic and contextual information. These embeddings serve as the input features to the classifier.

2.4.2 Training the Classifier

We trained a multiclass logistic regression model on the BERT embeddings, whose labels correspond to seven mental health categories outlined earlier. We then split the data into training and test sets using an 80-20 stratified split to maintain class balance. The classifier then models the relationship between the semantic features captured by BERT and the probability that a given statement belongs to each mental health class. For each input statement, the classifier outputs a probability distribution over all classes and assigns the label with the highest probability.

2.5 Baseline Model 3: K-Means Clustering Only

As a third baseline, we explored the utility of geometric clustering as a feature extraction method using the K-Means algorithm. While LDA focuses on probabilistic topic distributions, K-Means partitions the data into distinct, non-overlapping subgroups based on feature similarity (typically Euclidean distance). This approach tests whether the mental health statements form distinct geometric clusters in the vector space that correlate with their assigned labels, effectively evaluating if unsupervised grouping can serve as a proxy for supervised classification.

2.5.1 Cluster Feature Generation

We utilized the K-Means clustering algorithm to generate features for the downstream classifier. The algorithm iteratively partitions the dataset into k clusters by minimizing the variance within each cluster. For this baseline, the feature set X consists of the cluster representations derived from this process. Rather than using the raw text or dense embeddings directly, each data point is represented by its relationship to the learned cluster centroids. This transforms the high-dimensional text data into a lower-dimensional representation based purely on geometric proximity.

2.5.2 Training the Classifier

Similar to the previous baselines, we trained a multiclass Logistic Regression classifier on the extracted cluster features. We employed an 80-20 train-test split, stratified by the target variable to ensure that the distribution of the seven mental health statuses remained consistent across subsets.

To mitigate the impact of class imbalance, particularly for minority classes like *Personality Disorder* and *Stress*, the classifier was initialized with balanced class weights. The model was trained with a maximum of 5,000 iterations to facilitate convergence. By using Logistic Regression on top of K-Means features, we assess the linear separability of the unsupervised clusters with respect to the ground-truth mental health categories.

2.6 Hybrid Model 1: LDA + BERT

The first hybrid model augments BERT embeddings with topic-distribution features derived from Latent Dirichlet Allocation (LDA). After standard text preprocessing, LDA is applied to the corpus to generate a probability distribution over K latent topics for each document. These topic-proportion vectors are then concatenated with BERT embeddings to form a unified feature representation:

$$\mathbf{h} = [\text{BERT}_{1\dots d} \parallel \text{LDA}_{1\dots K}].$$

This combined representation allows the classifier to incorporate both contextual semantics and topic-level structure when predicting mental health categories.

2.7 Hybrid Model 2: TF-IDF + K-Means + BERT

The second hybrid model incorporates clustering-based structure instead of topic distributions. This approach is motivated by the short and sparse nature of many statements in the dataset, where traditional topic models may not perform optimally.

2.7.1 TF-IDF Vectorization

Each cleaned document is transformed into a TF-IDF vector. TF-IDF (Term Frequency–Inverse Document Frequency) quantifies the importance of a term within a document relative to its importance across the entire corpus. It is computed as:

$$\text{TF-IDF}(w, d) = \text{TF}(w, d) \times \text{IDF}(w),$$

where $\text{TF}(w, d)$ is the frequency of word w in document d , and $\text{IDF}(w)$ is higher for words that are rare across all documents.

This representation emphasizes words that are both frequent within a specific document and informative across the corpus. For example, in the statements:

- “I feel anxious and my chest is tight,”
- “Anxiety keeps me awake at night,”

words such as *anxious*, *chest*, and *awake* receive higher TF-IDF values than common terms like *feel*, *and*, or *me*. This makes TF-IDF particularly effective for short texts, where a few key terms often carry most of the semantic signal.

2.7.2 K-Means Clustering

K-Means clustering is applied to the TF-IDF matrix to identify groups of statements with similar lexical patterns. Each document receives a cluster assignment, which is then represented as a one-hot encoded vector or cluster-indicator feature. These cluster features summarize coarse-grained similarity structure across the corpus without relying on contextual embeddings.

2.7.3 Feature Fusion

For each document, the cluster-derived feature vector is concatenated with its BERT embedding:

$$\mathbf{h}' = [\text{BERT}_{1\dots d} \parallel \text{Cluster}_{1\dots C}],$$

where C denotes the number of clusters. This hybrid representation combines contextual meaning from BERT with lexical grouping information identified through clustering.

2.7.4 Classification

The fused vectors are used to train a Logistic Regression model to predict the seven mental health categories. This model serves as an alternative hybrid strategy for comparing clustering-based structure with topic-based structure in conjunction with transformer embeddings.

3 Results

3.1 Classification Reports

Class	Precision	Recall	F1-Score	Support
Anxiety	0.24	0.43	0.31	724
Bipolar	0.20	0.38	0.26	500
Depression	0.43	0.10	0.16	3017
Normal	0.75	0.43	0.54	3208
Personality Disorder	0.03	0.31	0.06	179
Stress	0.10	0.25	0.15	459
Suicidal	0.41	0.52	0.46	2128
Accuracy		0.34		10215
Macro Avg	0.31	0.35	0.28	10215
Weighted Avg	0.48	0.34	0.36	10215

Table 1: Classification Report for Baseline Model 1 (LDA Only)

Class	Precision	Recall	F1-Score	Support
Anxiety	0.68	0.80	0.74	724
Bipolar	0.57	0.75	0.65	500
Depression	0.74	0.49	0.59	3017
Normal	0.93	0.89	0.91	3208
Personality Disorder	0.21	0.69	0.33	179
Stress	0.40	0.69	0.51	459
Suicidal	0.63	0.66	0.64	2128
Accuracy		0.70		10215
Macro Avg	0.59	0.71	0.62	10215
Weighted Avg	0.74	0.70	0.71	10215

Table 2: Classification Report for Baseline Model 2 (BERT only)

Class	Precision	Recall	F1-Score	Support
Anxiety	1.00	0.07	0.13	724
Bipolar	0.00	0.00	0.00	500
Depression	0.00	0.00	0.00	3017
Normal	0.50	0.93	0.66	3208
Personality Disorder	0.03	0.55	0.06	179
Stress	0.00	0.00	0.00	459
Suicidal	0.50	0.23	0.31	2128
Accuracy		0.36		10215
Macro Avg	0.29	0.25	0.16	10215
Weighted Avg	0.33	0.36	0.28	10215

Table 3: Classification Report for Baseline Model 3 (K-Means Only)

Class	Precision	Recall	F1-Score	Support
Anxiety	0.68	0.80	0.74	724
Bipolar	0.56	0.75	0.64	500
Depression	0.75	0.49	0.59	3017
Normal	0.93	0.89	0.91	3208
Personality Disorder	0.22	0.70	0.33	179
Stress	0.40	0.70	0.51	459
Suicidal	0.63	0.67	0.65	2128
Accuracy		0.70		10215
Macro Avg	0.60	0.72	0.62	10215
Weighted Avg	0.74	0.70	0.71	10215

Table 4: Classification Report for Hybrid Model 1 (LDA + BERT)

Class	Precision	Recall	F1-Score	Support
Anxiety	0.69	0.80	0.74	724
Bipolar	0.56	0.76	0.65	500
Depression	0.74	0.49	0.59	3017
Normal	0.93	0.90	0.91	3208
Personality Disorder	0.21	0.68	0.32	179
Stress	0.40	0.69	0.51	459
Suicidal	0.63	0.66	0.65	2128
Accuracy		0.7002		10215
Macro Avg	0.59	0.71	0.62	10215
Weighted Avg	0.74	0.70	0.71	10215

Table 5: Classification Report for Hybrid Model 2 (Clustering + BERT)

3.2 Comparison of Baseline Models

Comparing the three baseline approaches reveals significant disparities between unsupervised feature extraction (LDA, K-Means) and supervised semantic embeddings (BERT).

3.2.1 Baseline 1 (LDA) vs. Baseline 3 (K-Means)

While Baseline 1 (LDA) and Baseline 3 (K-Means) achieve similar overall accuracy (0.34 vs. 0.36), their class-level performance differs fundamentally.

Class	LDA	K-Means	Difference
Anxiety	0.31	0.13	-0.18
Bipolar	0.26	0.00	-0.26
Depression	0.16	0.00	-0.16
Normal	0.54	0.66	+0.12
Personality Disorder	0.06	0.06	0.00
Stress	0.15	0.00	-0.15
Suicidal	0.46	0.31	-0.15

Table 6: Class-level differences in F1 Scores between LDA and K-Means

- **Class Collapse in K-Means:** Baseline 3 (K-Means) exhibits severe class collapse, failing to identify a single instance of Bipolar, Depression, or Stress (F1-score of 0.00). It achieves a slightly higher overall accuracy only by overfitting to the majority Normal class (Recall 0.93), which is reflected in the positive difference (+0.12) for that class.
- **Distribution in LDA:** Baseline 1 (LDA), despite lower accuracy, maintains non-zero F1-scores across all categories. It successfully captures some signal for Bipolar (0.26) and Depression (0.16), suggesting that topic distributions preserve more distinct class-specific information than simple geometric clustering in the absence of semantic embeddings.

3.2.2 Unsupervised Baselines vs. Baseline 2 (BERT)

Baseline 2 (BERT) vastly outperforms both LDA and K-Means, doubling the accuracy to 0.70. This confirms that for mental health text classification, deep semantic context is far more critical than latent topic distribution or geometric grouping alone.

Class	LDA	BERT	Difference
Anxiety	0.31	0.74	+0.43
Bipolar	0.26	0.65	+0.39
Depression	0.16	0.59	+0.43
Normal	0.54	0.91	+0.37
Personality Disorder	0.06	0.33	+0.27
Stress	0.15	0.51	+0.36
Suicidal	0.46	0.64	+0.18

Table 7: Class-level differences in F1 Scores between LDA and BERT

As shown in the table above, the transition from probabilistic topics (LDA) to contextual embeddings (BERT) yields massive improvements across the board, particularly for Anxiety and Depression (+0.43), which require nuanced understanding of sentiment rather than just keyword presence.

3.3 Impact of Hybridization

To evaluate the contribution of the transformer component, we compare each unsupervised baseline against its corresponding hybrid counterpart.

3.3.1 Baseline Model 1 vs. Hybrid Model 1

Integrating BERT embeddings with LDA topic features results in a substantial performance leap, increasing accuracy from 0.34 to 0.70.

Class	LDA Only	Hybrid 1	Difference
Anxiety	0.31	0.74	+0.43
Bipolar	0.26	0.64	+0.38
Depression	0.16	0.59	+0.43
Normal	0.54	0.91	+0.37
Personality Disorder	0.06	0.33	+0.27
Stress	0.15	0.51	+0.36
Suicidal	0.46	0.65	+0.19

Table 8: Class-level differences in F1 Scores between LDA and LDA + BERT

The hybrid model resolves the limitations of LDA by providing the semantic nuance required to distinguish between classes that share similar topics

(e.g., Depression and Suicidal), effectively doubling the F1-score for critical classes like Anxiety and Depression.

3.3.2 Baseline Model 3 vs. Hybrid Model 2

The addition of BERT to the K-Means baseline rescues the model from the class collapse observed in the unsupervised setting.

Class	K-Means	Hybrid 2	Difference
Anxiety	0.13	0.74	+0.61
Bipolar	0.00	0.65	+0.65
Depression	0.00	0.59	+0.59
Normal	0.66	0.91	+0.25
Personality Disorder	0.06	0.32	+0.26
Stress	0.00	0.51	+0.51
Suicidal	0.31	0.65	+0.34

Table 9: Class-level differences in F1 Scores between K-Means and Clustering + BERT

Most notably, the hybrid model restores the F1 scores for Bipolar, Depression, and Stress from 0.00 to competitive levels (0.51–0.65). This indicates that while geometric clusters alone are insufficient to separate these classes, they do not hinder the powerful semantic representations provided by BERT when concatenated.

3.3.3 Baseline Model 2 vs. Hybrid Model 1

Between the logistic regression model using BERT only and the hybrid model using BERT and LDA, the accuracy remains the same at 0.70. Similarly, the weighted averages for precision, recall, and F1 score are nearly identical across all classes. The following table shows the class level differences in the F1 score:

Class	BERT Only	LDA + BERT	Difference
Anxiety	0.74	0.74	0.00
Bipolar	0.65	0.64	-0.01
Depression	0.59	0.59	0.00
Normal	0.91	0.91	0.00
Personality Disorder	0.33	0.33	0.00
Stress	0.51	0.51	0.00
Suicidal	0.64	0.65	+0.01

Table 10: Class-level differences in F1 Scores between BERT and LDA + BERT

Additional Observations:

- **Slight class-level differences:**
 - F1-score for Suicidal increases slightly (+0.01) in the hybrid model.
 - F1-score for Bipolar decreases slightly (-0.01).
 - All other classes show no change in F1-score.
- **Minor improvement in macro-average recall:** Macro-average recall increases slightly from 0.71 to 0.72, suggesting a small gain in performance across less frequent classes.

3.3.4 Baseline Model 2 vs. Hybrid Model 2

Both models (BERT only and Clustering + BERT) achieve nearly identical overall accuracy, with BERT Only at 0.70 and Clustering + BERT at 0.7002, indicating minimal overall performance improvement. Additionally, the weighted metrics for precision, recall, and F1-score remain essentially unchanged, measuring at 0.74 for precision, 0.70 for recall, and 0.71 for F1. The following table illustrates the class-level differences in F1 scores between the two models:

Class	BERT Only	Clustering + BERT	Difference
Anxiety	0.74	0.74	0.00
Bipolar	0.65	0.65	0.00
Depression	0.59	0.59	0.00
Normal	0.91	0.91	0.00
Personality Disorder	0.33	0.32	-0.01
Stress	0.51	0.51	0.00
Suicidal	0.64	0.65	+0.01

Table 11: Class-level differences in F1 Scores between BERT and Clustering + BERT

Additional Observations:

- **Class-level differences:**

- F1-score for Suicidal slightly improves (+0.01) in the hybrid model.
- F1-score for Personality Disorder slightly decreases (-0.01).
- Other classes show no change in F1-score.

- **Minor improvement in macro-average recall:** Macro-average recall remains almost unchanged at 0.71 for both models.

- **Interpretability benefit:** By incorporating clustering-based topic structure, the hybrid model can potentially provide more interpretable insights into latent patterns in mental health text.

Conclusion: While the hybrid Clustering + BERT model shows minimal numerical improvements over the baseline, it provides an alternative hybrid approach that captures topic-level structure, which could be useful for interpreting patterns in less frequent or nuanced mental health categories.

3.3.5 Hybrid Model 1 vs. Hybrid Model 2

Comparing the two hybrid models, the LDA + BERT model (Hybrid Model 1) and the Clustering + BERT model (Hybrid Model 2), the overall accuracy remains nearly identical, with 0.70 for Hybrid Model 1 and 0.7002 for Hybrid Model 2. Weighted averages for precision, recall, and F1 score are also very

similar across all classes. The following table shows the class-level differences in F1 scores between the two hybrid models:

Class	LDA + BERT	Clustering + BERT	Difference
Anxiety	0.74	0.74	0.00
Bipolar	0.64	0.65	+0.01
Depression	0.59	0.59	0.00
Normal	0.91	0.91	0.00
Personality Disorder	0.33	0.32	-0.01
Stress	0.51	0.51	0.00
Suicidal	0.65	0.65	0.00

Table 12: Class-level differences in F1 Scores between LDA + BERT and Clustering + BERT

Additional Observations:

- The F1 scores for most classes remain unchanged between the two models, indicating very similar performance.
- Slight improvements are observed for the Bipolar class (+0.01) in the Clustering + BERT model, while Personality Disorder sees a minor decrease (-0.01).
- Overall, both hybrid approaches provide comparable accuracy and weighted metrics, suggesting that either topic extraction method (LDA or Clustering) can be effectively combined with BERT embeddings.

3.4 Confusion Matrices

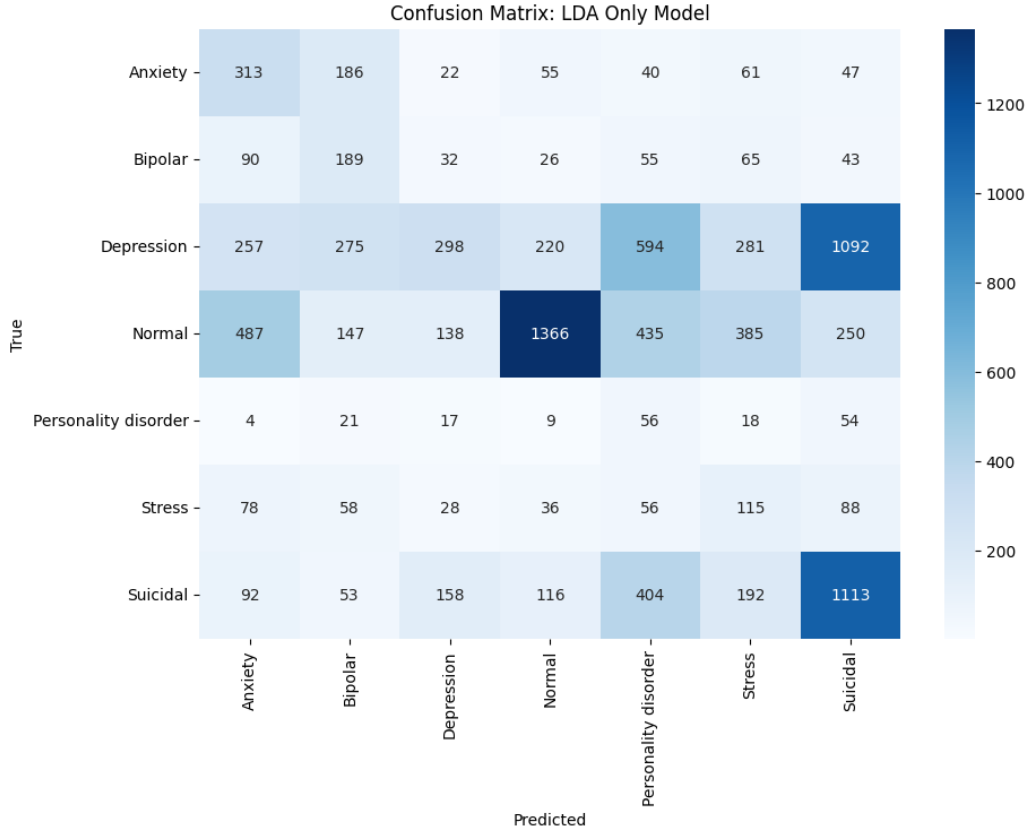


Figure 1: LDA-only Confusion Matrix

Figure 1 presents the confusion matrix for the LDA-only model, which relies solely on topic-distribution features extracted from TF-IDF text. As expected for a model without contextual understanding, overall performance is limited, with high rates of misclassification across nearly all mental-health categories.

The model performs best on the most frequent label, Normal, achieving 1366 correct predictions. However, even this decent result is paired with substantial confusion in other categories, particularly Anxiety (436 instances), Personality Disorder (435 instances), and Stress (385 instances). This sug-

gests that LDA captures broad thematic structure but struggles to distinguish everyday emotional language from clinically significant content.

Performance drops sharply for psychologically nuanced classes. Depression posts, for example, are widely dispersed across the matrix, with large numbers mislabeled as Suicidal (1092 cases) and Personality Disorder (594 cases). Similarly, Suicidal posts are often predicted as Depression (158) or Personality Disorder (404), reflecting LDA’s inability to separate different forms of high-intensity negative affect. Because topic models emphasize word co-occurrence rather than sentence-level meaning, emotionally charged categories collapse into a single broad cluster of distress.

Low-frequency and conceptually specific classes, such as Bipolar and Personality Disorder, are especially challenging, with Bipolar correctly identified only 189 times and frequently confused with Anxiety, Depression, and Personality Disorder, while Personality Disorder fares even worse, achieving just 56 true positives and being widely misclassified into Normal, Suicidal, and Depression. These patterns highlight LDA’s fundamental limitation: clinical distinctions often rely on subtle contextual cues, causal reasoning, and narrative framing that topic models cannot capture, underscoring that while LDA detects coarse lexical themes, it lacks the semantic precision required for mental-health text classification and motivating the subsequent use of BERT-based models, which incorporate contextual relationships to provide far more accurate psychological language representations.

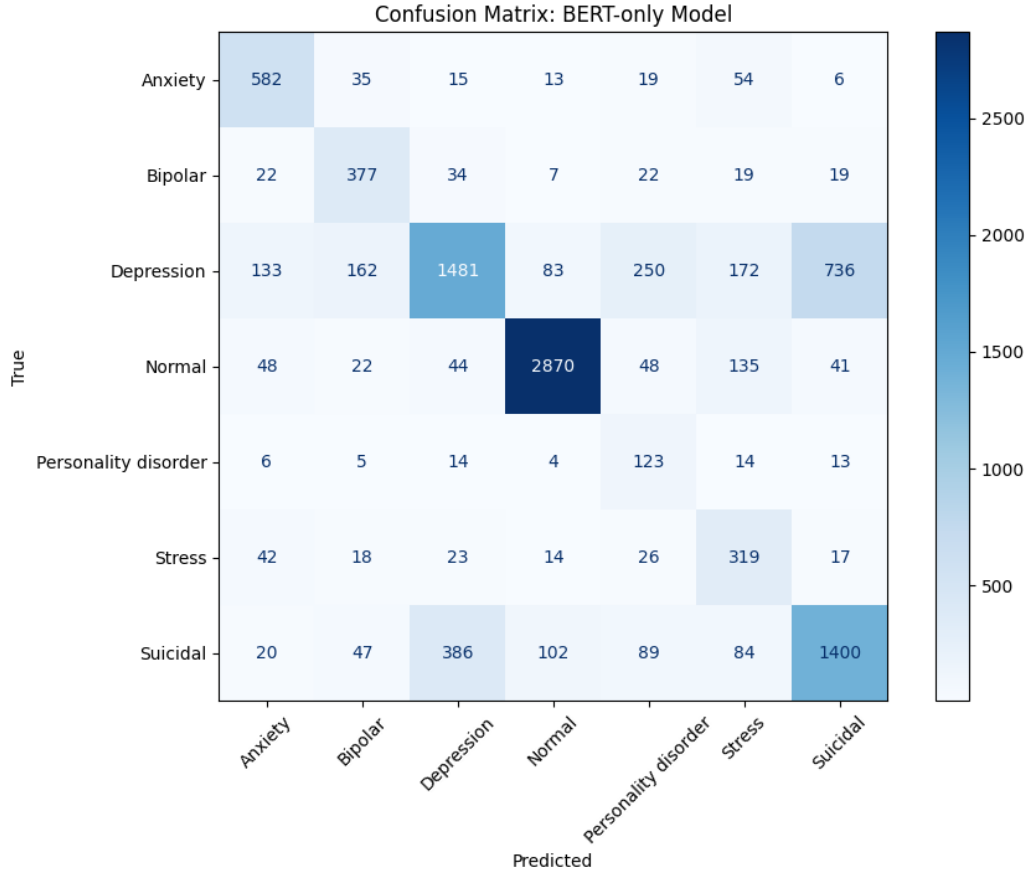


Figure 2: BERT-only Confusion Matrix

Figure 2 depicts the confusion matrix for our BERT baseline model. This model demonstrates strong overall performance, particularly for high-frequency categories. The model correctly identifies 2870 Normal statements, indicating excellent recognition of non-pathological text. Similarly, substantial true positives are observed for Depression (1481) and Suicidal ideation (1400), reflecting BERT’s ability to capture contextual distinctions in emotionally charged language.

However, the confusion matrix also highlights meaningful patterns of misclassifications. Many Depression posts are mislabeled as Suicidal (736 instances) and Personality Disorder (250 instances), illustrating the conceptual and linguistic overlap in expressions of despair and hopelessness. Likewise,

Anxiety and Stress frequently overlap, with Anxiety texts misclassified as Stress 54 times and Stress being assigned to Anxiety 42 times, likely due to shared vocabulary centered around physical tension, worry, and lack of control.

More subtle classes, such as Personality Disorders, show weaker performance, with only 123 correct predictions. This suggests that BERT struggles when the defining linguistic cues are subtle, psychologically technical, or spread across longer narrative descriptions. Overall, the confusion matrix confirms that BERT excels at modeling contextual signals in mental health data, but still encounters difficulty where categories share emotional semantics or when posts lack sufficient detail.

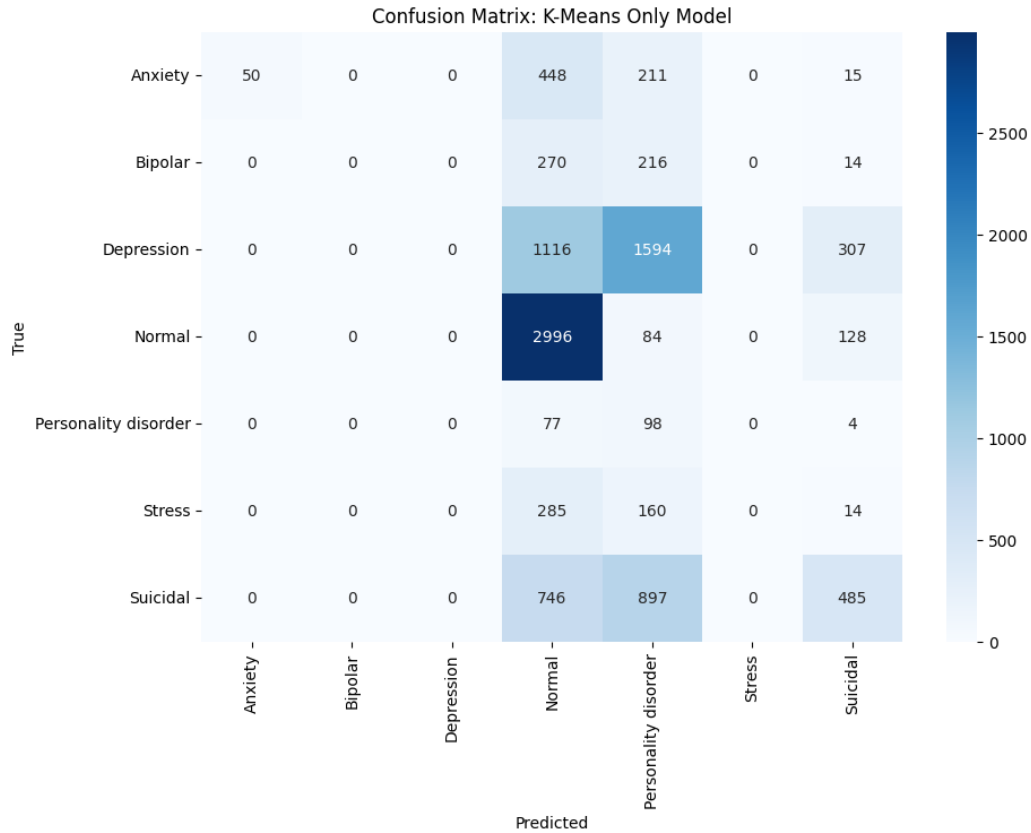


Figure 3: K-Means-Only Confusion Matrix

Figure 3 shows the confusion matrix for the K-means-only model, which only relies on cluster assignments derived from the TF-IDF text. Much like our LDA-only model, We can see that performance is extremely limited: nearly all predictions collapse into just two labels: Normal and Personality Disorder. While the model correctly identifies 2996 Normal posts, it also misclassifies large portions of every other category as Normal, including 1116 Depression posts, 285 Stress posts, and 746 Suicidal posts, indicating that the clusters fail to capture meaningful clinical distinctions.

A similar trend occurs with Personality Disorder, which absorbs high-intensity emotional content regardless of the true label. For instance, 1594 Depression posts and 897 Suicidal posts are predicted as Personality Disorder, reflecting how K-Means groups texts based on superficial lexical similarity rather than psychological nuance. Low-frequency classes such as Bipolar and Personality Disorder receive almost no correct predictions, further underscoring the model’s inability to distinguish subtle mental-health categories.

Overall, the K-Means confusion matrix demonstrates that unsupervised lexical clusters provide minimal discriminative value on their own. Without contextual or semantic information, the model collapses diverse emotional expressions into a few broad groups, reinforcing the need for more expressive representations such as BERT embeddings or hybrid models.

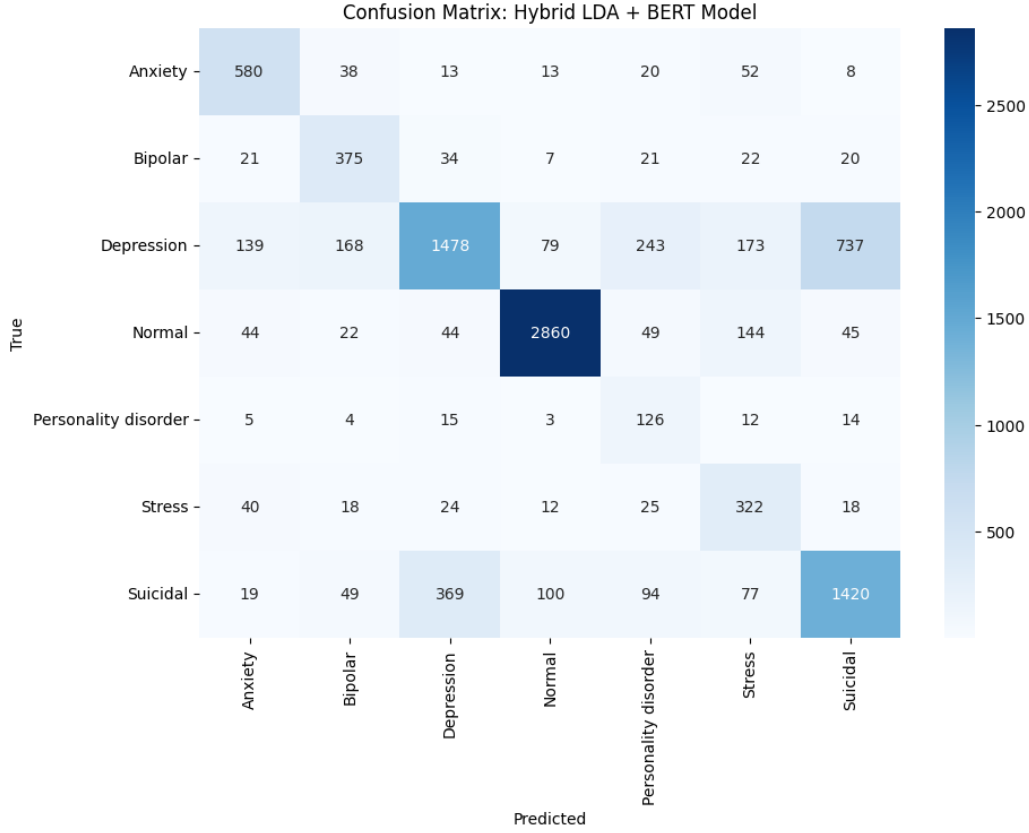


Figure 4: LDA and BERT Hybrid Confusion Matrix

Figure 4 depicts the confusion matrix for our first hybrid model, which involves LDA topic features that have been concatenated with BERT embeddings. This model’s confusion matrix is highly similar to the baseline BERT model shown in Figure 1, with marginal performance differences. True positives for major classes are almost identical; for example, we have 2860 correctly classified Normal posts, 1478 for Depression, and 1420 for Suicidal ideation, compared to 2870, 1481, and 1400 for the same categories in our BERT-only confusion matrix. Slight fluctuations in error distribution appear, such as Depression still being misclassified as Suicidal at a high rate (737 cases) and Anxiety again spilling into Stress (52 cases). These similarities demonstrate that the introduction of LDA topics provides little new predictive signal.

The small shifts seen in specific classes also support the idea that LDA might actually introduce more noise than benefit. For instance, Personality Disorder examples receive 126 correct predictions, nearly identical to the BERT baseline (123), but misclassifications into Depression (15) and Suicidal (14) persist. Short posts in this dataset often lack stable word co-occurrence structure for topic modeling, meaning LDA proportions may not meaningfully separate overlapping emotional categories. Because BERT already captures sequential and contextual information, LDA primarily adds redundant features. Hybrid Model 1 therefore increases interpretability, since topic features can be inspected independently, but does not enhance classification accuracy.

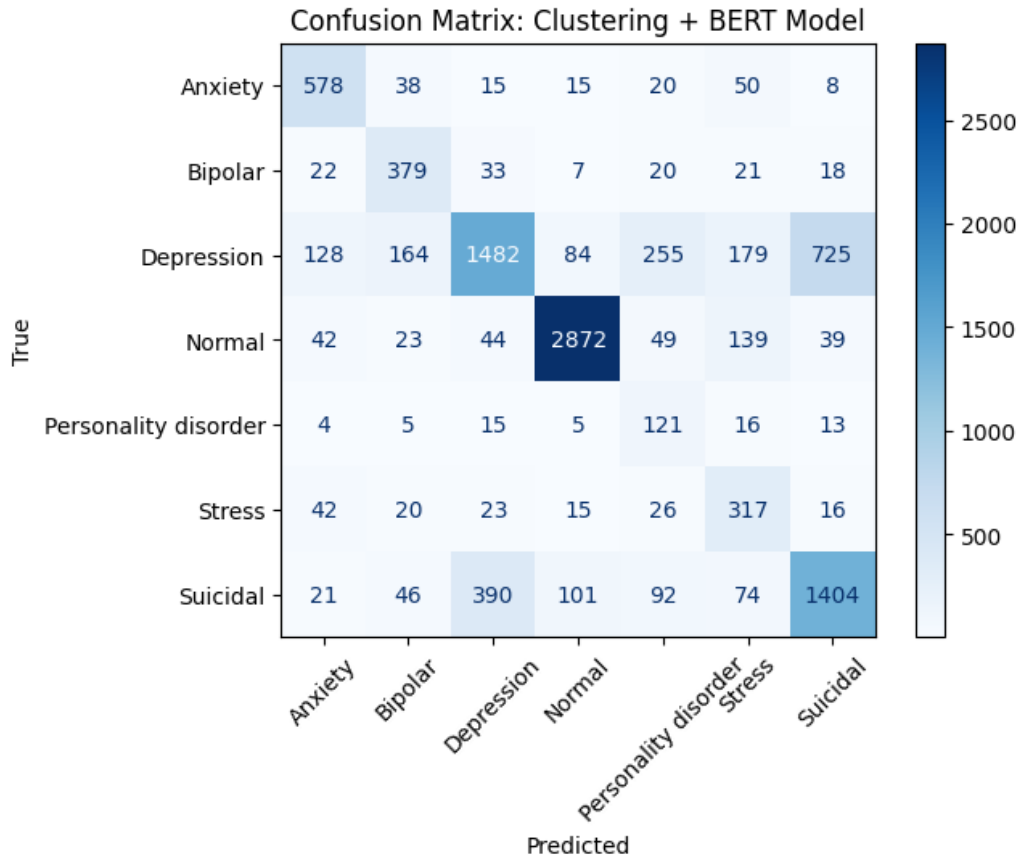


Figure 5: Clustering and BERT Hybrid Confusion Matrix

Figure 5 depicts the confusion matrix for our last model, Hybrid Model 2, which introduces cluster assignments derived from TF-IDF and K-Means to the BERT model, but the resulting confusion matrix again remains highly similar to the BERT-only baseline. The classifier maintains strong performance for major classes, including 2872 correct predictions for Normal, 1482 for Depression, and 1404 for Suicidal, indicating that the inclusion of lexical cluster identifiers does not disrupt BERT’s effectiveness. Misclassification relationships persist as well, such as Depression being predicted as Suicidal 725 times, reflecting the same conceptual proximity observed in the other models. Similarly, Anxiety and Stress continue to bleed into one another, with 50 Anxiety posts predicted as Stress and 42 Stress posts predicted as Anxiety, showing that lexical similarity alone cannot fully resolve emotional nuance.

The clustering features appear to slightly improve stability for certain classes. For example, Personality Disorder obtains 121 correct predictions, which is marginally higher than in the other models. However, the overall confusion profile continues to be dominated by overlapping linguistic markers, such as Bipolar being confused with Anxiety (38 cases) or Depression and Suicidal ideation co-occurring in model outputs. This indicates that while TF-IDF clusters capture coarse vocabulary patterns, they offer limited additional discriminative value beyond the contextual embeddings already encoded by BERT. The matrix therefore supports the conclusion that mental-health categories rely heavily on contextual cues—such as temporal framing, psychological reasoning, and emotional progression—that clustering methods cannot independently capture.

4 Discussion

4.1 Model Strengths and Weaknesses

4.1.1 Unsupervised Baselines (LDA and K-Means)

Baseline Model 1 (LDA) and Baseline Model 3 (K-Means) revealed the fundamental limitations of relying on statistical co-occurrence or geometric clustering for mental health classification.

- **LDA (Weakness):** While LDA provided interpretable topic distributions, it lacked the semantic resolution to distinguish between emo-

tionally overlapping categories. For example, Depression and Suicidal ideation share many high-probability words (e.g., "life," "pain," "tired"), causing the model to conflate them.

- **K-Means (Weakness):** K-Means failed significantly due to "class collapse," where the geometric centroids of minority classes (like Bipolar and Stress) were indistinguishable from the noise of broader categories. This suggests that without semantic embedding, mental health states do not form linearly separable clusters in TF-IDF space.

These models highlight that purely unsupervised feature extraction is insufficient for high-stakes classification tasks where labels depend on nuance rather than keyword frequency.

4.1.2 Baseline Model 2 (BERT Only)

The baseline model, which relies solely on BERT embeddings and a Logistic Regression classifier, demonstrates strong overall performance, achieving an accuracy of 0.70 and high F1-scores for dominant classes such as Normal, Anxiety, and Suicidal. Its primary strength lies in BERT's ability to capture deep contextual and semantic information from short mental-health statements, which is crucial in domains where subtle differences in phrasing can imply different emotional or psychological states.

However, the model exhibits some weaknesses, as seen in its relatively low F1 scores for minority classes like Personality Disorder and Stress, where the scores are 0.33 and 0.51, respectively. This indicates that the embeddings alone do not sufficiently distinguish these categories, likely due to limited representation of these classes in the dataset and the high overlap in linguistic patterns across similar mental-health conditions. In other words, BERT offers rich semantic information, but cannot overcome insufficient data or skewed distributions toward high-frequency classes without balancing techniques or task-specific fine-tuning. While BERT embeddings optimize general semantic similarity, they may miss fine-grained clinical distinctions. Subtle emotional cues such as intent, intensity, chronicity, and symptom-specific markers may not be effectively captured in BERT's fixed sentence representation, particularly with a general-purpose model that lacks fine-tuning on psychiatric or clinical text.

4.1.3 Hybrid Model 1 (BERT + LDA)

Hybrid Model 1 combines BERT embeddings with LDA-derived topic distributions with the goal of incorporating both semantic and thematic information. Although its quantitative performance is nearly identical to the baseline, the model’s main strength lies in the added interpretability provided by LDA. The topic proportions offer a transparent view of the broader themes present in mental-health statements, complementing BERT’s dense and less interpretable sentence-level representations. This can be useful when qualitative insight or model explainability is important.

That being said, the limited improvements in performance also highlight this model’s weaknesses. Because the dataset consists largely of short, sparsely worded statements, LDA often produces diffuse or poorly differentiated topics. As a result, the topic vectors lack the discriminative detail needed to meaningfully shift the decision boundaries learned from BERT embeddings. In many cases, different mental-health categories share similar high-frequency words (e.g., feel, tired, overwhelmed, can’t), leading LDA to assign similar topic proportions to statements that belong to different classes. This reduces the usefulness of the added features, since they do not reliably encode distinctions that the classifier can exploit.

In addition, LDA’s bag-of-words assumptions limit its ability to capture contextual or sequential cues that are often crucial in mental-health language; for example, distinguishing “I can’t calm down” (anxiety-related) from “I can’t go on” (suicidal ideation). BERT already captures these contextual nuances through its attention-based architecture, so the LDA features tend to provide redundant or lower-quality information relative to the embeddings. When these topic proportions are concatenated with BERT vectors, they contribute little new signal and may even introduce noise, which helps explain why the hybrid model does not outperform the baseline.

Overall, Hybrid Model 1 improves interpretability but not predictive accuracy. The combination of short input texts, overlapping vocabulary across classes, and LDA’s limited representational capacity means that topic-modeling features are not well aligned with the structure of this dataset or with the strengths of transformer-based sentence embeddings.

4.1.4 Hybrid Model 2 (Clustering + BERT)

Hybrid Model 2 replaces topic modeling with TF-IDF-based K-Means clustering, aiming to capture lexical similarity patterns that may not be fully emphasized in BERT’s contextual embeddings. Instead of relying on a probabilistic topic structure, this approach groups documents based on shared surface-level vocabulary, allowing the model to encode coarse-grained lexical patterns that might correlate with certain mental-health conditions. For short texts, such as those in this dataset, TF-IDF can elevate discriminative keywords (e.g., anxious, awake, tightness) that play a significant role in distinguishing statements but may contribute only weakly to dense BERT representations. These cluster assignments provide an additional structural layer that is often more flexible than traditional topic modeling and can be examined directly for interpretability or exploratory analysis.

Despite these advantages, the inclusion of cluster-based features does not lead to substantial performance improvements. This suggests that BERT embeddings alone already encode most of the discriminative information necessary for classification, leaving limited room for clustering to provide useful complementary signals. Additionally, K-Means imposes hard cluster boundaries that may not align with the nuanced, overlapping nature of mental-health categories. For instance, many psychological states share vocabulary and emotional tone, making it difficult for a purely lexical clustering method to form groups that correspond meaningfully to the target labels. The method is also sensitive to the chosen number of clusters C , so inappropriate values of C can introduce noise rather than clarify structure.

In summary, Hybrid Model 2 offers a more adaptable and potentially more useful structural representation than LDA, particularly for exploratory analysis or model interpretability. However, its contribution to predictive accuracy is negligible, indicating that lexical clustering does not add substantial new information beyond what is already captured by transformer-based sentence embeddings.

4.2 Computational Considerations

Both hybrid models rely on dense contextual embeddings generated by a pre-trained Sentence-BERT model, which is more efficient than full transformer fine-tuning but still computationally demanding. Generating BERT embeddings for the entire dataset can require nontrivial memory and pro-

cessing time, especially when concatenated with additional features such as LDA topic distributions or TF-IDF cluster indicators. Training the LDA model adds further computational overhead, since topic inference must be performed on every document/statement. The TF-IDF + K-Means approach is generally lighter than LDA, but it still involves building a high-dimensional term-document matrix and performing unsupervised clustering, which can be expensive for large corpora. Despite these constraints, both of the hybrid approaches were feasible to train on a standard CPU environment with needing GPU acceleration, suggesting they are practical for moderate-sized datasets and real-world mental health applications.

4.3 Ethical Considerations

Analyzing and classifying mental-health related text raises important ethical concerns. Misclassification can have serious implications if systems are used in clinical or support contexts. For instance, labeling a suicidal statement as normal could prevent timely interaction, while over-predicting severe conditions could unnecessarily cause stress for users or escalate situations. The dataset consists of anonymized social media posts, which introduces concerns about consent, privacy, and whether the data adequately represents diverse populations and cultural expressions of mental health. There is also the risk of embedded biases: models may learn patterns influenced by linguistic style, socioeconomic factors or demographic disparities. These models should be treated as supportive tools rather than diagnostic instruments, while clearly communicating the limitations of the model. Future work should include fairness evaluations and explore the system’s performance across different demographic groups to ensure responsibility and equity.

4.4 Limitations

Although the hybrid models combine lexical structure with contextual semantics, there are still several limitations present. Many statements in the dataset are short, informal, and noisy, which restricts the effectiveness of models like LDA that rely on richer document context to infer stable topics. This may reduce the benefit of adding LDA features to BERT embeddings. Similarly, while Logistic Regression is easy to interpret, it may not capture complex nonlinear interactions between features, potentially limiting predictive performance. The dataset may also contain class imbalance and concep-

tual overlap between categories such as Anxiety vs. Stress or Depression vs. Suicidal, making classification more challenging. Additionally, the lack of demographic metadata prevents evaluation of whether the models generalize across diverse populations. Finally, the models rely solely on text and do not incorporate other potentially informative signals, such as posting behavior, temporal patterns, or multimodal cues that could enhance real-world mental health assessment.

4.4.1 Impact of Post Length Variation

The dataset includes posts ranging from short, single-sentence statements to multi-paragraph narratives, and this variation influences several components of the modeling pipeline. LDA relies on sufficiently long documents to infer stable topic distributions, and very short posts often yield noisy or uninformative topic vectors. TF-IDF behaves differently for short and long texts, potentially leading to uneven cluster assignments during K-Means clustering. Although BERT handles short text effectively, longer posts may contain multiple emotional cues that are compressed into a single embedding. These disparities in document length introduce variability into the hybrid feature representations and may contribute to the hybrid models' limited performance gains relative to BERT alone.

5 Future Work

There are several promising directions for extending this work and addressing its current limitations. One key area is improving interpretability, especially for models that rely heavily on contextual embeddings. While LDA provides interpretable topic-level insights, transformer-based models can still be largely black-box systems. More research could explore methods such as Integrated Gradients, SHAP explanations for embeddings, or attention based interpretability mechanisms to provide clearer insight into which linguistic cues particularly drive predictions. Additionally, we could look at more advanced topic-aware frameworks such as BERTopic or Top2Vec which may produce higher quality, more stable topics for short texts, which could strengthen the connection of supervised structure and contextual embeddings.

Another direction of improvement could be incorporating multimodal data. Mental health expression online often extends beyond plain text and

includes images, emojis, audio snippets, and posting behavior metadata. Using textural models with image-based emotion detectors or speech based sentiment models could provide richer and more holistic understanding of a user’s psychological state. Multimodal fusion models might be capable of detecting warning patterns that the text alone cannot capture.

Future work could also focus on developing real-time or interactive applications. A real-time classifier could support triage systems for mental health support platforms, automatically surface concerning posts, or personalize responses in conversational agents. Achieving this would require efficiency improvements, GPU-optimized embeddings, and lightweight model distillation techniques to reduce inference time. Streaming data frameworks or on-device embeddings (e.g., using MiniLM or DistilBERT) might also be necessary for preserving privacy and scalable deployment.

Future iterations of this work should incorporate fairness and robustness evaluations. Expanding the dataset to include more diverse linguistic styles, dialects, and demographic contexts would help assess generalizability. Data augmentation techniques, longitudinal modeling of user behavior, or semi-supervised learning approaches could also strengthen model performance on underrepresented mental health categories. These extensions would move the model closer to real-world applicability while ensuring that it remains transparent, equitable, and clinically responsible.

6 Conclusion

In this study, we benchmarked unsupervised, supervised, and hybrid approaches for classifying mental health status from social media text. While traditional topic modeling (LDA) and clustering (K-Means) offer interpretability, they suffer from significant class collapse when used in isolation. Deep learning classifiers, specifically BERT, proved essential for capturing the semantic nuance of mental health language, achieving an accuracy of 0.70 compared to approximately 0.35 for the unsupervised baselines.

However, the addition of topic-based features to BERT did not further enhance predictive performance. Our results suggest that for short, informal mental health text, complex hybrid architectures may be over-engineering; a well-tuned transformer (BERT) is sufficient and more efficient than trying to force-fit topic modeling features. Consequently, future deployment efforts should prioritize the efficiency and explainability of transformer models

rather than the integration of lower-resolution text mining signals.

7 Code and Data Availability

Data and code scripts are available at <https://github.com/jordanandrew/NLP-Final-Project/tree/main>.

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A Top Three Statements Per Class Based on BERT Embeddings

A.1 Anxiety

1. been feeling feverish hello for a while now i have been feeling feverish but whenever i go get my temperature checked it always comes out as normal also i have been having these cramps in my legs arms and feet they are not exactly in extreme pain i guess but i feel that there is something terribly wrong is there something wrong is it just my anxiety acting up again i am also worried about having a brain tumor but then again the only proof i have is having some sort of brain fog i have been stumbling with my words and have trouble retrieving them it might be my anxiety can not stop thinking about this (prob=1.00)
2. restless restless anxious please be anxious (prob=1.00)
3. ha strikes again so i am not looking for reassurance or diagnosis or anything really i guess but i just had to share with some people who might understand i have a long history of health anxiety can not get a cold without convincing myself i might be dieing frequent rabies scares i have convinced myself something was seriously wrong a couple times a month for years i have mostly talked myself out of seeing a doctor because i know its just anxiety and i do not want doctors to think i am crazy or wasting their time a few days ago though i found a lump under the skin above my right rib cage so i did the logical thing and poked around for a lump on my left side i thought maybe i felt a smaller similar lump and talked myself off the ledge but now i can actually see a lump on my right side in the mirror and when i prod around it feels like there might be a cluster of small lumps only one is visible though so of course i am worried it might be cancer although dr google tells me there is all sorts of benign explanations i am currently talking myself into seeing a doctor for peace of mind but i hate the doubt telling me this is anxiety and ill be wasting the doctors time i am terrified the doctor will label me with hypochondria i do not know why i am posting this i am just frustrated that i can be terrified of illness and doctors at the same time anxiety is torture that worms into everything and just makes me feel so confused (prob=1.00)

A.2 Bipolar

1. been lurking over a year on this sub and wanted to share some good news so i have bipolar 1 with psychotic features as per my diagnoses and i wanted to share some good news with this wonderful community i will preface this by saying that after i got out of the hospital over a year ago pretty long and arduous stay around 4 months i was lost and i had no optimism about my future once i got my phone back i started googling med related things and this subreddit would always come up i started to realize slowly that there are a large number of people with bipolar living productive and stable lives and that exhibit linear thinking i thought i was relegated to being acutely ill for the rest of my life and that i would have at least two major manic episodes a year that would always get me hospitalized i will spare some of the details about these episodes because they are honestly a bit embarrassing and i imagine they could be triggering drug usage anyways it has been over a year since my last manic episode and i have not even experienced hypomania in the interim over the last year i went from living in group homes and care facilities to finally going back to college and getting as and went back to work i have been living in my own apartment for 5 months and have never had a period of stability this long i think what actually worked for me was staying on medication for more than 6 months and working with a treatment team i am actually really happy with my psychiatrist from stanford and actually appreciate his style he does not just shove pills in my face and actually takes over an hour to ask about how i am doing and then during the last 5 minutes we talk about meds he was even willing to take me off of my monthly injection and switch me to a lower dose oral dosage which seemed shocked me as the shot was the one thing that i would consistently take due to noncompliance this made me realize that he trusted me and after that i started trusting him my relationships are really solid now too and i have actually kept a friend instead of burning all my bridges like before i actually got a job volunteering at one of the residential facilities i was a client at and this really helps to give me a feeling of fulfillment i also started making music again which really helps as well lastly i think that staying sober with the exception of the occasional beer with family has been a huge factor in my stability tldr humble bragged a little got medicated got sober turned life around and really appreciate

that something like this community exists to know we are not alone (prob=1.00)

2. how does lamictal help for you guys hi guys i have just recently discovered that i believe i have bipolar ii disorder and i have started the process of getting treated i have made an appointment with a psychiatrist which i will be seeing next month and in the meantime i have been researching medications one particular one i have been interested in is lamictal it seems perfect for me since i suffer more from bipolar depression as opposed to mania plus it does not seem to have as worrisome side effects as lithium for those of you on lamictal what kind of symptoms did you have in order for your doctor to prescribe it and does it work well for you (prob=1.00)
3. just diagnosed looking for some help with this i was recently diagnosed and put on lamictal looking back on life i realize how bipolar has made me act towards certain aspects and has really turned my life upside down i do not know how to recognize the episodes or what to do even when i do recognize one what is the best thing for you how do you know when you are going through a depressive or high episode (prob=1.00)

A.3 Depression

1. wer mit einem depressiven menschen zu tun hat st t unweigerlich an seine grenzen denn der umgang mit ihnen ist anstrengend in great depression schreibt unsere kolumnistin ber psychische erkrankungen http t co pb wfwnglf (prob=0.99)
2. the back road a tweet on depression in le than 0 character http t co hi oucfa m (prob=0.98)
3. depression is the result of an unnatural overdevelopment of civilisation to the point where our minds have been forced to become complacent due to lack of any real purpose in the days of survival humans would wake up everyday faced with a viariety of challenges that would need to be overcome in order to simply stay alive theory of the depression pandemic that affects so many of us today (prob=0.98)

A.4 Normal

1. that is correct (prob=1.00)
2. that must be nice (prob=1.00)
3. then i have to refer to (prob=1.00)

A.5 Personality Disorder

1. is it possible to have both avoidantdismissive attachment style and avpd do any of you have both i read that when you have avoidantdismissive attachment style you have often good selfconfidence and that is why you can not have avpd is that true i think they are two different things and while you may feel like you can do things best yourself it does not mean you can not feel shame an inferior to others (prob=1.00)
2. do avpd make you deny or suppress parts of yourself also can you mention what parts it suppresses view poll (prob=1.00)
3. what is avpd like i had social anxiety or still have some of it i also was diagnosed with avpd by one therapist but i am not sure if that one fits i tick a few boxes and every personality style test says i am anxious avoidant but i do not really know what that would mean so what it something that makes someone avpd in your experience (prob=1.00)

A.6 Stress

1. he thinks i have too much stress (prob=1.00)
2. no stress stress is present in everyone is life this is unavoidable what we can change is how we deal with it and sometimes all we need is to know how to relieve stress quickly before it does damage to us or our loved ones so simple and quick ways for you to relieve your stress they are suitable for any situation whether at work in studies at home in a daytoday situation or in something more punctual good reading chat (prob=1.00)

3. facing stress at the workplace for the past two months i was struggling to give my 100 and was not satisfied with my work in the office thankfully i found a blog online about tips to deal with work stress and it helped me in getting rid of stress (prob=1.00)

A.7 Suicidal

1. i want to kill myself i am so fucking done (prob=0.99)
2. i am done next time i fuck up i will kill myself by slitting both my wrists i do not want to live on this planet anymore i hope i fuck up soon so this all can end my suicide plans (prob=0.99)
3. the last thing i want is a failed attempt short and guaranteed ways to kill myself (prob=0.99)